

# Irrationality after One Stabilization of Universally $CH_0$ -Trivial Cubic Threefolds

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## Abstract

Does universal  $CH_0$ -triviality force rationality after stabilization? We construct a non-isotrivial one-parameter family of smooth cubic threefolds  $X_b$  for which  $X_b$  is universally  $CH_0$ -trivial but  $X_b \times \mathbf{P}^1$  is irrational. More generally, we prove that  $X \times \mathbf{P}^1$  is irrational for every smooth cubic threefold  $X$ .

The cycle argument proves an all-degree integral saturation theorem for marked finite-etale graph quotients of elliptic powers: divided powers of the marked divisor lattice are ordinary divisor products. The six-axis intermediate Jacobians in the nonstandard  $A_5$ -invariant cubic pencil satisfy its hypotheses. Voisin's theorem gives universal  $CH_0$ -triviality on every smooth fibre.

For irrationality, we count primitive-sixth eigenvalues of framed formal monodromy on the numerical small even quantum connection. Divisor tags make Iritani's blowup formula compatible with a possibly noninjective center specialization. The count is a birational invariant through dimension four and equals two for a cubic. Thus the cycle-theoretic and quantum detectors diverge on the same smooth fourfolds.

**Keywords.** cubic threefold; universal  $CH_0$ -triviality; irrationality; intermediate Jacobian; quantum connection; weak factorization.

## 1 Introduction

A smooth projective variety is universally  $CH_0$ -trivial if the degree map

$$\deg : CH_0(Y_L) \longrightarrow \mathbf{Z}$$

is an isomorphism for every field extension  $L/\mathbf{C}$ . This condition is a central consequence of stable rationality, and its failure is a standard stable-rationality obstruction. Its validity, however, need not provide a rational parametrization. We ask how far that gap can persist after one explicit stabilization.

Our examples are fourfolds of the form  $Y = X \times \mathbf{P}^1$ , where  $X$  is a smooth cubic threefold in the nonstandard  $A_5$ -invariant pencil. Two independent mechanisms act on the same  $Y$ . The intermediate Jacobian of  $X$  has an algebraic primitive minimal class; this gives universal  $CH_0$ -triviality. The small even quantum connection of every smooth cubic has primitive-sixth framed formal monodromy; Iritani's quantum blowup formula shows that its multiplicity survives one stabilization.

**Theorem 1.1** (Separation after one stabilization). *Let  $W_5$  be the five-dimensional irreducible  $A_5$ -module. Inside*

$$\mathbf{P}((\mathrm{Sym}^3 W_5^*)^{A_5}).$$

let  $B^\circ$  be the smooth locus. For the resulting family  $\mathcal{X} \rightarrow B^\circ$  and every  $b \in B^\circ(\mathbf{C})$ ,

$$\mathcal{X}_b \text{ is universally } CH_0\text{-trivial,} \quad \mathcal{X}_b \times \mathbf{P}^1 \text{ is universally } CH_0\text{-trivial and irrational.}$$

The family is non-isotrivial in coarse moduli.

The irrationality statement does not depend on the symmetry of the pencil.

**Theorem 1.2** (One-step irrationality). *For every smooth complex cubic threefold  $X$ , the fourfold  $X \times \mathbf{P}^1$  is irrational.*

The same invariant also gives one-step irrationality for every smooth prime Fano threefold of genus eight; see Corollary 4.12.

The invariant cubic space in Theorem 1.1 is two-dimensional. Indeed, the character of  $W_5$  is  $(5, 1, -1, 0, 0)$  on the classes  $1, 2A, 3A, 5A, 5B$ . The symmetric-cube character formula gives

$$\dim(\mathrm{Sym}^3 W_5^*)^{A_5} = 2.$$

Thus its projectivization is a pencil with connected smooth locus  $B^\circ$ . Hartlieb denotes it by  $M_{H_1}$ ; it is not the component associated with  $\mathbf{1} \oplus W_4$  [9, Lemma 5.5].

We do not claim that  $X$  is stably irrational. From the second stabilization onward, weak-factorization centers can themselves have cubic dimension, and the low-dimensional exclusion used here no longer applies.

## How the two proofs meet

The cycle argument starts from six elliptic quotients of the Fano surface. Their Rosati Gram matrix is  $6I - J$ , and the geometric kernel lies in the exotic two-point complement of  $\mathbf{P}^1(\mathbf{F}_2) \subset \mathbf{P}^1(\mathbf{F}_4)$ . Its local graph slopes are finite etale at two and three. The graph equations then make the descended divisor lattices rank-one generated. Since rank-one coefficient divisors have square zero,  $\Theta^4/4!$  is an ordinary integral divisor product.

The quantum argument counts primitive sixth-root eigenvalues of framed formal monodromy on the numerical small even quantum connection. Iritani's operation formulas make this count additive through weak factorization; divisor tags repair a potentially noninjective center specialization. Cai's cubic calculation is nonzero, while every center of dimension at most two contributes zero. The special  $A_5$ -geometry enters only in the cycle argument; the quantum theorem applies to every smooth cubic threefold.

## Relation with earlier rationality results

Clemens and Griffiths proved that every smooth complex cubic threefold is irrational by means of its intermediate Jacobian [5]. That obstruction does not by itself control a product with projective space, and stable rationality of smooth cubic threefolds remains open. Cai recently recovered irrationality from the primitive-sixth-root formal monodromy of the cubic quantum connection [4]. Iritani and Iritani–Koto prove the quantum  $D$ -module decompositions for blowups and projective bundles [10, 12]; Katzarkov–Kontsevich–Pantev–Yu identify the nef-canonical normal form used for low-dimensional centers [13]. Theorem 1.2 combines these inputs at the next dimension.

On the cycle side, Voisin reduces universal  $CH_0$ -triviality of a cubic threefold to algebraicity of the primitive minimal class of its intermediate Jacobian [18]. Beckmann–de Gaay Fortman place such minimal classes in a general integral-Fourier framework [3]. For the very general cubic threefold the minimal class is not algebraic: every curve class on the intermediate Jacobian is an even multiple of it, so a very general cubic threefold admits

no decomposition of the diagonal [7, Theorem 1.3 and Corollary 1.4]. What Voisin's criterion leaves is the countable union of moduli subvarieties on which the class is algebraic; Theorem 3.5 places an explicit exceptional family inside that union. Here the six elliptic quotients supplied by Roulleau and the special period locus isolated by Hartlieb make the class explicit: here it belongs to the ordinary integral divisor-product lattice for every smooth member of the nonstandard  $A_5$ -pencil.

## Proof map

Section 2 computes the six-axis polarization and identifies the exotic cubic gluing. Section 3 proves the local divisor-product theorem and applies it to the primitive minimal class. Section 4 proves Theorem 1.2 from formal monodromy and weak factorization. The final section combines the two detectors and records the exact scope boundary.

## 2 The six-axis polarization and the exotic packet

Let  $\Omega$  be the six conjugate  $D_5$ -subgroups of  $A_5$ , equivalently the six Sylow-5 subgroups with their normalizers, and put

$$H_2 = \text{Aug}(\mathbf{F}_2^\Omega)/\langle \mathbf{1} \rangle.$$

The module  $H_2$  is simple and  $\text{End}_{A_5}(H_2) = \mathbf{F}_4$ .

The six axes arise from the Fano surface. Let  $S$  be the Fano surface of a smooth member of the  $A_5$ -pencil and  $(J, \Theta) = \text{Alb}(S)$  its intermediate Jacobian. Each of the six  $D_5$ -subgroups gives an elliptic quotient  $q_H : J \rightarrow E_H$  and its Rosati dual  $i_H : E_H \rightarrow J$ .

**Proposition 2.1** (Six-axis polarization). *After coherent identification of the elliptic factors, the Rosati Gram matrix of the six maps is*

$$G_6 = 6I_6 - J_6.$$

*Any five axes therefore give an isogeny  $E^5 \rightarrow J$  whose pulled-back polarization has Smith type  $(1, 6, 6, 6, 6)$ .*

*Proof.* We recall the numerical passage from Roulleau's genus-two configuration to the six elliptic fibres. Let  $D_g \subset S$  be the curve indexed by an involution  $g \in A_5$ , put  $D_{\text{tot}} = \sum_g D_g$ , and, for a dihedral subgroup  $H$  of order ten, put

$$F_H = \sum_{g \in H, g^2=1} D_g.$$

Roulleau proves that  $F_H$  is a fibre of  $S \rightarrow E_H$  [17, Theorem 11(D)], so  $F_H^2 = 0$ . His intersection table has  $D_g^2 = -4$ , and distinct curves meet with multiplicity two or one when the product of their labels has order three or five, respectively. Among the 105 unordered pairs of involutions, the product orders two, three, and five occur 15, 30, 60 times. Hence

$$D_{\text{tot}}^2 = 15(-4) + 2(30 \cdot 2 + 60 \cdot 1) = 180.$$

Every involution belongs to two of the six dihedral subgroups, and therefore  $\sum_H F_H = 2D_{\text{tot}}$ . Two-transitivity on the six subgroups now gives

$$30(F_H \cdot F_{H'}) = (2D_{\text{tot}})^2 = 720, \quad F_H \cdot F_{H'} = 24 \quad (H \neq H').$$

After coherent identification of the generic non-CM elliptic quotients, two-transitivity makes the Rosati matrix have diagonal  $d$  and off-diagonal  $m$ . Put

$$D_H = q_H^*[0] \in \text{NS}(J), \quad N_H = i_H q_H \in \text{End}(J).$$

Then  $D_H|_S = F_H$ , and  $N_H$  is the Rosati endomorphism attached to  $D_H$ . Coherent identification gives  $q_H i_H = [d]$  and  $q_H i_{H'} = [m]$  for  $H \neq H'$ . The invariant sum of the six axis maps vanishes because the five-dimensional constituent has no trivial summand; composing with  $q_H$  gives  $d + 5m = 0$ . The minimal-class identity  $[S] = \Theta^3/3!$ , together with the polarized Riemann–Roch trace formula

$$\frac{\alpha\beta\Theta^{g-2}}{(g-2)!} = \text{tr}(N_\alpha) \text{tr}(N_\beta) - \text{tr}(N_\alpha N_\beta),$$

obtained by polarizing the reduced-norm Riemann–Roch polynomial [8, Theorem 4.1], gives

$$\text{tr}(N_H) = d, \quad \text{tr}(N_H N_{H'}) = m^2, \quad F_H F_{H'} = d^2 - m^2.$$

Thus  $d^2 - m^2 = 24$ . Together with  $d + 5m = 0$  and  $d > 0$ , this forces  $(d, m) = (5, -1)$ , hence  $G_6 = 6I - J$ . A five-by-five minor has Smith form  $(1, 6, 6, 6, 6)$ .  $\square$

Let  $B^\circ$  be the connected smooth locus of the nonstandard  $A_5$ -pencil, and let  $\mathcal{J} \rightarrow B^\circ$  be its relative intermediate Jacobian.

**Lemma 2.2** (Relative six-axis source). *There are an elliptic scheme  $\mathcal{E} \rightarrow B^\circ$  and a relative isogeny*

$$f : \mathcal{A} := \mathcal{E} \otimes_{\mathbf{Z}} (\mathbf{Z}^\Omega / \mathbf{Z}\mathbf{1}) \longrightarrow \mathcal{J}$$

*which is  $A_5$ -equivariant. The quotient lattice carries the form induced by  $6I_6 - J_6$ ; both the source and this form carry their natural  $S_6$ -action. Moreover,*

$$f^* \lambda_\Theta = \lambda_{\mathcal{A}}. \tag{2.1}$$

*After omitting one axis,  $\mathcal{A} \simeq \mathcal{E}^5$  and this form has matrix  $6I_5 - J_5$ . If  $\mathcal{V} = \mathcal{E}[2]$ , the two-primary kernel of the polarization on the source has a natural  $A_5$ -equivariant symplectic description*

$$\mathcal{D}_2 \simeq H_2 \otimes_{\mathbf{F}_2} \mathcal{V}. \tag{2.2}$$

*The two-primary part  $\mathcal{K}_2$  of  $\ker f$  is an  $A_5$ -stable relative maximal isotropic subgroup of  $\mathcal{D}_2$ .*

*Proof.* For  $H \in \Omega$ , the integral norm  $n_H = \sum_{h \in H} h$  is a relative endomorphism of  $\mathcal{J}$ . Its action on rational first homology is ten times the projector onto the  $H$ -fixed summand. Fibrewise, the representation-theoretic isogeny decomposition is the self-product of an elliptic curve through the five-dimensional module  $W_5$  [9, Remark 5.8]. Since  $\dim W_5^H = 1$ , the fixed summand has rank two. More concretely,  $n_H^2 = 10n_H$ . The scheme-theoretic image of a homomorphism of abelian schemes in characteristic zero is an abelian subscheme, and its quotient factorization commutes with base change [2, Theorem A]. Thus the scheme-theoretic image  $\mathcal{E}_H \subset \mathcal{J}$  is an abelian scheme of relative dimension one. It is therefore an elliptic scheme. Since  $hn_H = n_H$  for every  $h \in H$ , the subgroup  $H$  acts trivially on  $\mathcal{E}_H$ . Moreover  $N_{A_5}(H) = H$ , so transport by an element carrying  $H$  to  $H'$  is independent of the chosen transporter. The six elliptic schemes are therefore coherently isomorphic; fix one of them as  $\mathcal{E}$ .

Write  $i_H : \mathcal{E} \rightarrow \mathcal{J}$  for the transported primitive inclusions. On the geometric generic fibre these maps are the six vectors of the standard five-dimensional coefficient representation, whose sum is zero. The connected smooth curve  $B^\circ$  is integral,  $\mathcal{E}$  is integral, and  $\mathcal{J}$  is

separated. Equality on the generic fibre therefore gives  $\sum_H i_H = 0$  over  $B^\circ$ . Hence  $(i_H)_{H \in \Omega}$  descends from  $\mathcal{E}^\Omega$  to the displayed map  $f : \mathcal{A} \rightarrow \mathcal{J}$ , and the construction is  $A_5$ -equivariant.

The connected  $D_5$ -quotient used by Roulleau has Rosati-dual image equal to the unique  $H$ -fixed elliptic subvariety. Its connected kernel makes that dual inclusion primitive, so the quotient is, up to a target automorphism, the Rosati adjoint of  $\mathcal{E}_H \hookrightarrow \mathcal{J}$ . Consequently the fibre classes in Proposition 2.1 compute these relative homomorphisms, rather than merely isogenous quotients. On the generic non-CM fibre they give diagonal entry 5 and off-diagonal entry  $-1$ . Applying the same generic-density and separatedness argument to these homomorphisms gives the identities over  $B^\circ$ , hence (2.1). Choosing any five coordinate classes identifies the quotient lattice with  $\mathbf{Z}^5$ , where its matrix is  $6I_5 - J_5$ . Since this has Smith type  $(1, 6, 6, 6, 6)$ ,  $f$  is a finite flat isogeny and its kernel is maximal isotropic in the source polarization kernel.

The two-primary discriminant local system is

$$\mathcal{D}_2 = \ker(6I_6 - J_6 : \mathcal{A}[2] \longrightarrow \mathcal{A}^\vee[2]).$$

Modulo two, the radical of the induced coefficient form on  $(\mathbf{F}_2^\Omega)/\langle \mathbf{1} \rangle$  is canonically

$$\text{Aug}(\mathbf{F}_2^\Omega)/\langle \mathbf{1} \rangle = H_2.$$

Tensoring this constant coefficient module with the finite etale local system  $\mathcal{V} = \mathcal{E}[2]$  gives (2.2) as an isomorphism of local systems. The discriminant commutator pairing is the tensor product of the coefficient form with the Weil pairing on  $\mathcal{V}$ , so the isomorphism is symplectic. Since  $f$  is  $A_5$ -equivariant,  $\mathcal{K}_2$  is  $A_5$ -stable; its fibrewise order is  $2^4$ , so it is a relative maximal isotropic subgroup of this rank-eight local system.  $\square$

**Proposition 2.3** (Principal gluing packet). *The  $A_5$ -stable two-primary principal kernels of the six-axis source are parametrized by*

$$\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}.$$

*The three  $\mathbf{F}_2$ -rational kernels are  $S_6$ -stable. Each of the other two has stabilizer exactly  $A_5$ , and the outer normalizer exchanges them. The intermediate Jacobian of the nonstandard  $A_5$ -cubic component uses the latter pair. Every  $A_5$ -stable three-primary kernel is a scalar graph.*

*Proof.* At two the coefficient discriminant is the simple four-dimensional  $\mathbf{F}_2 A_5$ -module  $H_2$ , whose commutant is  $\mathbf{F}_4$ . The full discriminant is  $H_2 \otimes_{\mathbf{F}_2} \mathbf{F}_2^2$ . An  $A_5$ -stable simple half is therefore a one-dimensional  $\mathbf{F}_4$ -subspace of  $\mathbf{F}_4^2$ , and every such half is maximal isotropic for the tensor product of the coefficient form with the alternating multiplicity form. This gives  $\mathbf{P}^1(\mathbf{F}_4)$ .

Here is the polarization check explicitly. Identify the coefficient heart with  $\mathbf{F}_4^2$  and put

$$b(x, y) = \text{Tr}_{\mathbf{F}_4/\mathbf{F}_2} \det(x, y).$$

On two multiplicity copies use

$$\Omega((x, y), (x', y')) = b(x, y') + b(y, x').$$

Every scalar  $a \in \mathbf{F}_4$  is self-adjoint for  $b$ , since  $b(ax, y) = b(x, ay)$ . Thus the vertical line and all four graphs  $\Gamma_a$  are isotropic of half dimension. The five module-theoretic possibilities are exactly the five polarized gluings; no further quadratic condition removes the exotic pair.

The simplex group acts on both multiplicity copies of the coefficient heart over  $\mathbf{F}_2$ . Hence the three lines defined over  $\mathbf{F}_2$  retain  $S_6$ . The stabilizer of an exotic graph is equally intrinsic. A permutation that fixes  $\Gamma_\omega$  commutes with  $\omega$ , hence is  $\mathbf{F}_4$ -linear on the two-dimensional  $\mathbf{F}_4$ -space  $H_2$ . If its determinant is  $d \in \mathbf{F}_4^\times$ , preservation of  $b$  says

$$\mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(dz) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(z) \quad (z \in \mathbf{F}_4).$$

The trace pairing is nondegenerate, so  $d = 1$ . It therefore lies in  $\mathrm{SL}_2(4)$ , of order 60. The given  $A_5$  already has that order and lies in the stabilizer, proving equality. The odd coset in its order-120 normalizer is semilinear and applies field Frobenius, so it exchanges  $\omega$  and  $\omega^2$ .

To pass from the generic fibre to the family, put

$$\mathcal{P} = \mathbf{P}_{\mathbf{F}_4}(\mathbf{F}_4 \otimes_{\mathbf{F}_2} \mathcal{V}).$$

By Lemma 2.2, this is the five-sheet local system of  $A_5$ -stable principal halves. It has the degree-three sub-local system  $\mathcal{P}_{\mathrm{rat}} = \mathbf{P}_{\mathbf{F}_2}(\mathcal{V})$  and the degree-two complement  $\mathcal{P}_{\mathrm{ex}}$ . Here  $H_2$ , together with its commutant  $\mathbf{F}_4$ , is constant, while monodromy acts only on  $\mathcal{V}$ . Its action is therefore  $\mathbf{F}_4$ -linear, not semilinear, and factors through  $\mathrm{GL}_2(\mathbf{F}_2)$ ; in particular it preserves both subsets. The relative kernel  $\mathcal{K}_2$  gives a section of  $\mathcal{P}$ .

At the geometric generic point this section cannot be rational. Otherwise the natural faithful  $S_6$ -action on  $\mathcal{A} = \mathcal{E} \otimes (\mathbf{Z}^\Omega/\mathbf{Z}\mathbf{1})$ , which is the identity on  $\mathcal{E}$  tensored with the faithful permutation action on the quotient lattice, would preserve the rational kernel: under (2.2) it acts on  $H_2$  and trivially on  $\mathcal{V}$ , so every member of  $\mathcal{P}_{\mathrm{rat}}$  is  $S_6$ -stable. Since this action preserves  $6I_6 - J_6$ , it would descend through the principally polarized quotient. It would act faithfully: if a source automorphism acts trivially on the quotient, the image of its difference from the identity is both connected and contained in the finite isogeny kernel, hence zero. Strong Torelli gives an exact sequence

$$1 \longrightarrow \{\pm 1\} \longrightarrow \mathrm{Aut}(J, \Theta) \longrightarrow \mathrm{Aut}(X) \longrightarrow 1.$$

The kernel of the resulting map from  $S_6$  to  $\mathrm{Aut}(X)$  would be a normal subgroup of order at most two; it is trivial. This would inject  $S_6$  into the generic automorphism group  $A_5$ , a contradiction [9, Section 5]. The section is therefore generically in  $\mathcal{P}_{\mathrm{ex}}$ . Its inverse image is open and closed in the connected base  $B^\circ$ , so every smooth fibre has exotic two-primary kernel.

At three the coefficient heart  $H_3$  is simple and  $\mathrm{End}_{A_5}(H_3) = \mathbf{F}_3$ . Hence its  $A_5$ -stable halves are the four graphs parametrized by  $\mathbf{P}^1(\mathbf{F}_3)$ , all scalar in a suitable graph chart.  $\square$

For the cycle-theoretic argument, the unordered pair is enough. Both members act on the heart with the same squarefree minimal polynomial  $t^2 + t + 1$ . Theorem 3.5 uses only the conclusion that the geometric kernel is not one of the three  $\mathbf{F}_2$ -rational gluings. At three it uses only scalarity of the possible  $A_5$ -stable graphs, not the finer monodromy selection.

Hartlieb proves that its intermediate-Jacobian image is a one-dimensional special subvariety [9, Proposition 5.7]. Since the cubic period map is a locally closed embedding, the resulting smooth family is non-isotrivial. Only this landing and nonconstancy are used below; no identification of compactified parameter curves is needed for the separation theorem.

### 3 Integral divisor products and the primitive minimal class

Let  $(A, \Theta)$  be a principally polarized abelian variety of dimension  $g$ . Put

$$P_A^k = \text{im}\left(\text{Sym}^k \text{NS}(A) \longrightarrow H^{2k}(A, \mathbf{Z})\right), \quad \gamma_A = \frac{\Theta^{g-1}}{(g-1)!}.$$

The class  $D^k/k!$  of a divisor on an abelian variety is integral. The question here is whether these divided powers already belong to the smaller lattice  $P_A^k$  generated by ordinary products of integral divisors.

For quotients of elliptic powers, this question is controlled by the valuation pattern of the descended coefficient forms. The main local result of this section says that finite-etale graph slopes force all divided powers of the prescribed graph divisor lattice into ordinary products, in every degree. The proof has two steps: the graph equations produce a symmetric matrix-of-ideals lattice, and a midpoint inequality makes that lattice rank-one generated. Rank-one coefficient divisors have square zero, so their divided powers expand without factorials. The six-axis application supplies the primitive minimal class needed below.

#### 3.1 A local graph theorem

Work over a complete unramified discrete valuation ring  $O$ , with uniformizer  $p$ . Let an elliptic-power source have coefficient polarization

$$G = PB, \quad P = \bigoplus_a p^a I_{M_a}, \quad B = \bigoplus_a B_a, \quad (3.1)$$

where each  $B_a$  is unimodular. A *marked finite-etale graph presentation* consists of a ruling in which the principal kernel is the graph of an endomorphism  $T$ , with the following properties:  $T$  preserves the depth summands  $M_a$ ,

$$T^t B^{-1} = B^{-1} T \quad \text{or, equivalently,} \quad B T^t = T B, \quad (3.2)$$

and the algebra generated by  $T$  modulo  $p^a$  is finite etale on every positive-depth summand. Thus adjointness is taken for the dual coefficient form  $B^{-1}$ . No etaleness is required on  $M_0$ .

After a finite unramified extension splitting these etale algebras, the positive-depth summands decompose orthogonally into blocks

$$L_i = L_{a_i, \lambda_i}, \quad T_i = t_i I + p^{a_i} S_i, \quad (3.3)$$

with  $S_i$  integral. We keep the unit part as one block. Write  $\delta_{ij} = v_p(t_j - t_i)$ , with  $v_p(0) = \infty$ , for two positive-depth blocks.

**Lemma 3.1** (The graph coefficient lattice). *In the split basis, the coefficient lattice of descended graph divisors is the symmetric matrix-of-ideals lattice whose diagonal block  $i$  has depth  $p^{a_i}$  and whose  $(i, j)$ -block has depth*

$$e_{ij} = \max\{a_i, a_j, a_i + a_j - \delta_{ij}\}. \quad (3.4)$$

*For the unit block and a positive-depth block  $j$ , one has  $e_{0j} = a_j$ . In particular,*

$$e_{ij} \geq \max(a_i, a_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil. \quad (3.5)$$

*Proof.* In graph coordinates put

$$C = \begin{pmatrix} P^{-1} & P^{-1}T \\ 0 & I \end{pmatrix}.$$

For a symmetric source coefficient  $A$ , direct multiplication gives

$$C \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} C^t = \begin{pmatrix} P^{-1}(AT^t - TA)P^{-1} & P^{-1}A \\ -AP^{-1} & 0 \end{pmatrix}. \quad (3.6)$$

Thus descent is equivalent to the integrality of the three displayed nonzero blocks. Here  $A = A^t$  is a symmetric coefficient bilinear form in this normalization. On two positive-depth blocks  $L_i \times L_j$ , the two off-diagonal blocks first force  $A_{ij}$  to have depth  $\max(a_i, a_j)$ . Substituting  $T_i = t_i I + p^{a_i} S_i$  into the upper-left block leaves the additional condition

$$p^{-a_i - a_j}(t_j - t_i)A_{ij} \in M_{r_i \times r_j}(O);$$

the terms containing  $S_i, S_j$  are already integral. Intersecting the two conditions gives (3.4). Changing a scalar lift cannot change its maximum. Pairing the unsplit unit block with  $L_j$  gives  $e_{0j} = a_j$  directly from (3.6), and (3.5) follows.  $\square$

The inequality (3.5) is the valuation shadow of the tropical positive semidefinite cone studied by Yu [19]. We need the following signed integral lift of that inequality; in particular, no division by two is permitted.

**Lemma 3.2** (Rank-one generation over a DVR). *Let  $L$  be a symmetric matrix-of-ideals lattice over  $O$ , with diagonal depths  $p^{a_i}$  and cross depths  $p^{e_{ij}}$ . Then  $L$  is generated by its rank-one forms if and only if*

$$2e_{ij} \geq a_i + a_j \quad (i \neq j). \quad (3.7)$$

*Proof.* If  $cvv^t \in L$ , the two diagonal conditions imply

$$v_p(cv_i v_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil.$$

Hence rank-one forms cannot generate a shallower cross ideal.

Conversely, fix a generator  $p^e(uv^t + vu^t)$  of a cross block. Choose  $x \geq a_i$  with  $y = 2e - x \geq a_j$ . The integers  $x, y$  have the same parity. If  $t = \min(x, y)$ , write  $x = t + 2r$  and  $y = t + 2s$ . Then

$$p^t(p^r u + p^s v)(p^r u + p^s v)^t - p^x u u^t - p^y v v^t = p^e(uv^t + vu^t). \quad (3.8)$$

All three rank-one terms lie in  $L$ . Together with the diagonal rank-one forms, these identities generate  $L$ , without division by two.  $\square$

**Theorem 3.3** (All-degree graph saturation). *For a marked finite-etale graph presentation, every divided power of a class in its prescribed graph divisor lattice is an ordinary integral divisor product. Equivalently, after localization at  $p$ , the divided-power envelope of that lattice and its ordinary product image agree in every degree.*

*Proof.* After the unramified splitting extension, Lemmas 3.1 and 3.2 express every class in the extended graph divisor lattice as a finite sum  $D = R_1 + \cdots + R_m$  of rank-one coefficient classes. These are classes in the coefficient extension of the original lattice, not newly descended line bundles. On the elliptic-power source the alternating form of each  $R_i$  has



rank two, hence  $R_i^2 = 0$ . Pullback by the isogeny is injective on torsion-free cohomology, so the same identity holds on the quotient. Therefore

$$\frac{D^k}{k!} = \sum_{1 \leq i_1 < \dots < i_k \leq m} R_{i_1} \cdots R_{i_k}, \quad (3.9)$$

an  $O'$ -linear combination of ordinary products over the splitting ring  $O'$ . More precisely,

$$P_{\text{gr}}^k \otimes_O O' = \text{im} \left( \text{Sym}^k(N_{\text{gr}} \otimes_O O') \longrightarrow H^{2k} \otimes_O O' \right).$$

Faithful flatness descends the quotient of these finite lattices, and hence descends (3.9) to  $\mathbf{Z}_p$ .  $\square$

The theorem concerns the divisor lattice selected by the marked elliptic-power presentation. If  $E$  has no complex multiplication, this is the full Neron–Severi lattice. At a CM fibre extra divisor classes need not satisfy the graph hypotheses. The statement is cohomological: it neither constructs Chow divided powers nor asserts that every integral Hodge class is a divisor product.

For comparison with the classical rational theory, Milne proves that the Lefschetz classes are the rational algebra generated by divisors [15, Theorem 3.2]. Divided powers in Chow and integral Fourier transforms are studied in [16, 3]. The local result above instead keeps the weighted integral graph lattice and the ordinary, rather than divided, product image.

### 3.2 The six-axis packet

We now apply the graph theorem to the polarization computed in Section 2.

**Lemma 3.4** (Local chart for the six-axis quotient). *Let  $G = 6I_5 - J_5$ . At each prime  $p \mid 6$ , the coefficient lattice has an orthogonal decomposition*

$$(\mathbf{Z}_p^5, G) \simeq U_0 \perp pU_1,$$

where  $U_0$  has rank one and  $U_1$  rank four, both unimodular. The principal quotient is trivial on  $U_0$ . On  $U_1/pU_1$ , its graph slope has minimal polynomial  $t^2 + t + 1$  at  $p = 2$ ; at  $p = 3$  it is a scalar, hence a one-point finite-etale block.

*Proof.* Let  $e_1, \dots, e_5$  be the standard basis. Since  $\langle e_1, e_1 \rangle = 5$  and  $\langle e_1, e_j \rangle = -1$ , the vectors

$$v_j = e_j + \frac{1}{5}e_1 \quad (2 \leq j \leq 5)$$

span the orthogonal complement of  $e_1$  over  $\mathbf{Z}_p$ . Its Gram matrix is  $\frac{6}{5}(5I_4 - J_4)$ , and  $\det(5I_4 - J_4) = 5^3$ . This gives the asserted unit line and depth-one block at two and three.

At two, Proposition 2.3 identifies the geometric kernel with an exotic graph. Its slope is  $\omega \in \mathbf{F}_4$ , whose polynomial is  $t^2 + t + 1$ . At three the coefficient heart is simple with scalar commutant. These are the actual kernels of the six-axis quotient, so the local presentations persist throughout each connected smooth component.  $\square$

**Theorem 3.5** (Divided powers for the six-axis family). *For every smooth member  $X_b$  of the nonstandard  $A_5$ -cubic pencil, the prescribed six-axis graph divisor lattice on  $J_b$  has all its integral divided powers in the ordinary divisor-product image. In particular,*

$$\frac{\Theta_b^4}{4!} \in P_{J_b}^4. \quad (3.10)$$

*Proof.* Only the primes two and three occur in the Smith form  $(1, 6, 6, 6, 6)$ . Lemma 3.4 verifies the hypotheses of Theorem 3.3 at both primes. At every other prime the isogeny identifies the integral source and target lattices. Since the relevant cokernels are finitely generated, the local memberships give the integral assertion. The polarization itself belongs to the prescribed graph lattice, yielding (3.10). Extra Neron–Severi classes at a special fibre can only enlarge the ordinary product image needed for this last assertion.  $\square$

Voisin identifies the primitive minimal class as the exact obstruction needed for cubic threefolds.

**Corollary 3.6** (Universal  $CH_0$ -triviality). *Every smooth member of the nonstandard  $A_5$ -cubic pencil is universally  $CH_0$ -trivial.*

*Proof.* By Theorem 3.5, the primitive minimal class of the intermediate Jacobian is algebraic. Voisin’s Corollary 4.4 [18] says that, for a smooth complex cubic threefold, this is equivalent to universal  $CH_0$ -triviality.  $\square$

The argument is fibrewise. It constructs neither a horizontal minimal cycle over the pencil nor a relative decomposition of the diagonal. Neither is needed for Corollary 3.6.

## 4 Framed formal monodromy

The obstruction used here belongs to the small even quantum connection itself. We follow it through each blowup in a weak factorization, without passing to equivalence classes of quantum atoms. This avoids a subtlety: a formal bulk gauge is coefficientwise Laurent in the loop coordinate, but need not specialize to a meromorphic gauge at a nearby analytic point.

The proof has three stages. We first define the framed multiplicity  $\nu_6$ . We then prove its projective-bundle and blowup formulas, using divisor tags when the center Novikov map is not injective. Finally we show that centers of dimension at most two have  $\nu_6 = 0$ , so weak factorization makes  $\nu_6$  birationally invariant through dimension four.

Let  $T$  be a smooth projective variety. Write

$$\Lambda_T = \mathbf{C}[[\mathrm{NE}_{\mathrm{num}}^{\mathbf{Z}}(T)]]$$

for the completed monoid ring of the image of effective homology classes in the numerical curve lattice

$$N_1(T) = \mathrm{im}(H_2(T, \mathbf{Z}) \longrightarrow \mathrm{Hom}(\mathrm{NS}(T)/\mathrm{tors}, \mathbf{Z})).$$

Thus  $N_1(T)$  is torsion-free and is separated by integral divisors. Let  $\mathcal{K}_T$  be an algebraic closure of its fraction field. The loop coordinate  $z$  remains independent: in particular,  $\mathcal{K}_T$  contains no chosen root of  $z$ . Apply the Levelt–Turrittin construction to the small even quantum connection over  $\mathcal{K}_T((z))$ . Its formal monodromy is the action of one turn on the original  $z$ -disc. Thus it includes both regular-singular monodromy and the deck permutation of ramified exponential factors.

**Definition 4.1.** Let  $M_f(T)$  be this framed formal-monodromy operator, and put

$$\nu_6(T) = \mathrm{mult}_{e^{\pi i/3}} M_f(T) + \mathrm{mult}_{e^{-\pi i/3}} M_f(T).$$

The multiplicities are algebraic multiplicities in the characteristic polynomial.

The characteristic polynomial is unchanged by extending one algebraic closure of the numerical Novikov fraction field to a common overfield. Thus  $\nu_6(T)$  is independent of that choice.

The frame matters. Adjoining roots of Novikov monomials is harmless, but adjoining a root of  $z$  would replace the original loop and could erase the deck part of the operator.

**Definition 4.2** (Pro-Laurent framed monodromy). Let  $B$  be a complete separated multiplicatively filtered  $k$ -algebra, and let  $V$  be a finite-dimensional  $k$ -vector space. The pro-Laurent gauge group is

$$\mathcal{G}_{B,F}^{\text{pro}}(V) = \varprojlim_N \text{GL}(V \otimes (B/F^N B)((z))).$$

It is generally larger than  $\text{GL}(V \otimes B((z)))$ , because the lower Laurent bound may decrease with  $N$ . For a compatible differential-module system over the quotients  $B/F^N B$ , its framed-monodromy characteristic polynomial is the common finite-level polynomial, when these polynomials agree under reduction. A pro-Laurent gauge preserves it if every finite-level gauge uses only integral powers of  $z$ .

We first record two formal facts used in the comparison theorems. They also fix the completion convention in the proof.

**Lemma 4.3** (Coefficientwise base change). *Let  $D$  be a formal differential module over  $k((z))$ , and let  $B$  be a complete separated  $I$ -adic  $k$ -algebra on which  $d/dz$  is zero. Scalar extension in the coefficient direction preserves the characteristic polynomial of framed formal monodromy. The same holds after a connection gauge whose entries use only integral powers of  $z$ .*

*Proof.* Choose a Levelt–Turrittin solution algebra  $R$  for  $D$  with differential constants  $k$ . At every finite level, the constants of  $R \otimes_k B/I^N$  are  $B/I^N$ : expansion in a  $k$ -basis reduces this to  $R^{d/dz} = k$ . A fundamental solution matrix then identifies the horizontal module with

$$\text{Sol}(D) \otimes_k B/I^N.$$

Passing to the inverse limit gives  $\text{Sol}(D) \hat{\otimes}_k B$ , with monodromy  $M_f(D) \otimes 1$ . An integral- $z$  gauge is single-valued on the original disc and conjugates this action.  $\square$

**Lemma 4.4** (Formal base shift). *Suppose a pullback of the even quantum connection of  $T$  is defined over a complete separated multiplicatively filtered coefficient ring  $B$ , and its bulk coordinate has the form*

$$a_0 \mathbf{1} + a_2^\circ + \eta, \quad a_2^\circ \in H^2(T) \otimes F^0 B, \quad \eta \in F^1 B \hat{\otimes} H^{\text{ev}}(T).$$

*Assume that  $a_0$  and  $a_2^\circ$  are constant in the positive filtration. Its framed formal-monodromy characteristic polynomial equals that of the small connection after the divisor substitution*

$$Q^d \longmapsto e^{\langle a_2^\circ, d \rangle} Q^d. \tag{4.1}$$

*Proof.* The string equation removes  $a_0 \mathbf{1}$  by a scalar exponential in  $z^{-1}$ ; this is single-valued and has trivial framed monodromy. The divisor equation gives (4.1). It remains to remove  $\eta$ , including its component in  $H^2(T)$ .

Modulo  $F^N B$ , the flat bulk equations

$$\partial_{\eta_i} G = -z^{-1}(\phi_i \star_\eta) G, \quad G|_{\eta=0} = 1,$$

have an invertible solution with bounded Laurent order in  $z$ . These equations are compatible because the even quantum connection is flat: the mixed derivatives of the quantum-product matrices agree, and those matrices commute by commutativity and associativity. Coefficient recursion in total bulk degree therefore gives the unique normalized formal solution. After substituting  $\eta$ , a term of total bulk degree  $m$  lies in  $F^m B$ ; modulo  $F^N B$ , only the finitely many degrees  $m < N$  survive. The resulting finite sum is congruent to the identity modulo  $F^1 B$ , hence is invertible, and has a Laurent lower bound depending on  $N$ . Uniqueness makes these solutions compatible in  $N$ . Their inverse limit belongs to

$$\mathcal{G}_{B,F}^{\text{pro}}(H^{\text{ev}}(T)). \quad (4.1a)$$

At each finite quotient the gauge uses only integral powers of  $z$ , hence is single-valued on the original disc. Lemma 4.3 makes the characteristic polynomial constant there. Compatibility of the finite-level solutions identifies these polynomials under every reduction map, so the hypothesis in Definition 4.2 is satisfied and they define its pro-system invariant. No Levelt–Turrittin decomposition over a larger analytic field, and no evaluation at a nonnilpotent analytic bulk point, is used.  $\square$

We now apply Iritani’s quantum blowup theorem and the Iritani–Koto projective-bundle theorem. Both are stated over completed homological Novikov rings. Iritani’s subsequent notes record explicitly that the blowup change of variables and decomposition preserve cohomological parity [11, Section 2].

**Lemma 4.5** (Numerical Novikov base change). *The quotient from the effective homology monoid to its image in  $N_1(T)$  extends continuously to completed monoid rings, and the quantum connection and the two comparison theorems base-change along it.*

*Proof.* Filter both monoids by intersection with an ample divisor. Effective homology classes below a fixed cutoff form a finite set, as follows from the finite-type Chow varieties of cycles of bounded degree. The quotient is therefore defined on every truncated monoid ring and passes to the inverse limit. The quantum products descend by finite sums of coefficients with the same numerical class. Pushforward of curves, pairing with  $c_1(N_{C/T})$ , and the derivations  $Q\partial_Q$  depend only on numerical classes. Since the comparison theorems provide both a map and its inverse, they remain inverse after this base change.  $\square$

**Proposition 4.6** (Framed operation formulas). *Let  $V \rightarrow T$  be a vector bundle of rank  $r \geq 2$ . Then*

$$\nu_6(\mathbf{P}_T(V)) = r\nu_6(T). \quad (4.2)$$

*Let  $\tilde{T} = \text{Bl}_C T$ , where  $C \subset T$  is smooth of codimension  $c \geq 2$ . In Iritani’s comparison field one has*

$$\nu_6(\tilde{T}) = \nu_6(T) + \sum_{j=0}^{c-2} \nu_6(C; \chi_j), \quad (4.3)$$

*where  $\chi_j$  is the numerical Novikov specialization of the  $j$ -th center summand. The endpoint terms in (4.3) are the intrinsic small invariants; the center terms are computed after the displayed specializations.*

*Proof.* Iritani’s Theorem 5.18 gives a formal coordinate isomorphism and a quantum- $D$ -module isomorphism

$$\text{QDM}(\tilde{T}) \cong \text{QDM}(T) \oplus \text{QDM}(C)^{\oplus(c-1)}$$

over a Laurent extension in the exceptional Novikov variable [10, Theorem 5.18]. The isomorphism and its inverse use only integral powers of  $z$ , and the mirror coordinates are

independent of  $z$ . The comparison field adjoins  $q^{-1/s_c}$ , where  $s_c = c - 1$  for even  $c$  and  $s_c = 2(c - 1)$  for odd  $c$ ; this is only a coefficient extension. The reconstruction coordinate below is  $u = q^{-1/(c-1)}$ . The center base shift is controlled before passage to the final Laurent chart. Put  $u = q^{-1/(c-1)}$  and use the reconstruction coordinates of [10, Section 5.8.2]:

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = -(c - 1)\lambda_j + h_{C,j} + O(u). \quad (4.4)$$

For the  $j$ -th target summand, let  $R_j$  be the image of  $Q_C^d \mapsto Q^{i_*d} u^{\rho_C \cdot d}$ , together with  $u$  and  $s_j$ , and set

$$J_j = (u, \{Q^{i_*d} u^{\rho_C \cdot d} : d \neq 0\}, \text{components of } s_j) \subset R_j, \quad B_j = \varprojlim_N R_j / J_j^N.$$

The center connection is defined over this image ring by [10, Remark 5.6]. Formula (4.4), together with (5.45), (5.47), and the initial asymptotics (5.27)–(5.30), shows that after the unit term  $-(c - 1)\lambda_j$  and fixed divisor  $h_{C,j}$  are removed, its target bulk coordinate lies in  $J_j H^*(C)$ . The noninjective center monomial map merely replaces the source ring by its image.

First normalize the target connection over  $\mathbf{C}((u))$  by the single-valued string scalar for  $-(c - 1)\lambda_j$  and by the fixed divisor shift. The normalized connection descends to  $B_j$ . Apply Lemma 4.4 modulo every  $J_j^N$ , take the compatible inverse limit, and only then extend scalars to Iritani's Laurent comparison field and apply  $\Psi$ . The fixed-divisor substitution is a  $z$ -constant coefficient automorphism, so it does not change  $\nu_6$ ; it is well defined on the image because  $h_{C,j}$  is a scalar multiple of  $\rho_C$ . Thus each center target summand has the framed monodromy of its specialized small connection. The ambient target uses  $\tau = \tau^\circ + t$ , with  $\tau^\circ = O(q^{-1})$ , and the same finite-quotient argument. Direct-sum additivity gives (4.3).

The endpoint map is faithful: on numerical curve lattices

$$\bar{\beta} \longmapsto (\phi_* \bar{\beta}, -E \cdot \bar{\beta})$$

is injective because  $N^1(\tilde{T}) = \phi^* N^1(T) \oplus \mathbf{Z}[E]$ . Hence the Laurent extension preserves the intrinsic endpoint characteristic polynomials.

The projective-bundle proof is parallel, again in reconstruction coordinates. Its comparison field adjoins  $q^{-1/r'}$ , with  $r' = r$  when  $r - 1$  is even and  $r' = 2r$  when  $r - 1$  is odd; the reconstruction coordinate remains  $u = q^{-1/r}$ . The extra root is again a coefficient extension. Put  $u = q^{-1/r}$ . For its  $j$ -th target summand let  $R_j$  be the image of  $Q_T^d \mapsto Q^d u^{c_1(V) \cdot d}$ , together with  $u$  and  $s_j$ , and set

$$J_j = (u, \{Q^d u^{c_1(V) \cdot d} : d \neq 0\}, \text{components of } s_j) \subset R_j, \quad B_j = \varprojlim_N R_j / J_j^N.$$

Proposition 5.6 and Section 5.8 of Iritani–Koto give

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = r\lambda_j - \frac{2\pi i j}{r} c_1(V) + O(u).$$

This is Proposition 5.6 and Section 5.8, especially (5.11), of [12]. Normalize the Laurent unit and fixed divisor terms first. The normalized connection descends to  $B_j$ ; apply Lemma 4.4 in its finite quotients, and then extend scalars to the Laurent comparison field. Again the divisor substitution is a  $z$ -constant coefficient automorphism; it factors through the displayed  $u$ -exponent. This gives  $r$  copies of the intrinsic base module and proves (4.2). The base Novikov map is the embedding in [12, (1.1) and Remark 5.2]. The theorem

assumes  $V^\vee$  globally generated. Tensoring  $V$  by a sufficiently negative line bundle achieves this without changing  $\mathbf{P}_T(V)$  [12, Remark 1.2]; the resulting divisor shift preserves framed monodromy by Lemma 4.4. All other Novikov extensions are in the coefficient direction. After the odd bulk variables vanish, pullback, pushforward, Fourier transformation, and multiplication by the characteristic-class factors all preserve cohomological parity. Hence both decompositions restrict to the even quantum connections used here. Formula (4.2) is tautological when  $r = 1$ .  $\square$

The center map in (4.3) is not necessarily injective [10, (5.15)]. Thus generic vanishing for  $C$  does not by itself imply that its specialized term vanishes. The next lemma supplies the missing implication.

**Definition 4.7.** A monomial map  $\chi : \Lambda_T \rightarrow A$  is *strictly Novikov-admissible* if  $A$  is a complete separated valued domain with domain associated graded, every monomial has nonzero image, and

$$v(\chi(Q^d)) = \ell(d), \quad (4.5)$$

where  $\ell$  is positive and proper on the effective numerical monoid.

We write  $\nu_6(T; \chi)$  for the framed small-even invariant after scalar extension along such a map.

The center maps in (4.3) are strictly admissible. Give the exceptional Laurent variable valuation zero and use an ample divisor  $H$  on  $T$ . Then

$$v_H(Q^{i_*d} q^{-\rho_C \cdot d/(c-1)}) = (H|_C) \cdot d.$$

The restriction  $H|_C$  is ample, which gives positivity and properness; the associated graded Laurent monoid ring is a domain.

**Lemma 4.8** (Divisor tagging). *Let  $\chi : \Lambda_T \rightarrow A$  be strictly Novikov-admissible. If the intrinsic small even quantum connection of  $T$  has no primitive-sixth framed monodromy, neither does its specialization along  $\chi$ .*

*Proof.* Choose integral divisors  $D_1, \dots, D_\rho$  whose pairings separate  $N_1(T)$ , and introduce formal variables  $\mathbf{t} = (t_1, \dots, t_\rho)$ . Define

$$\chi_{\mathbf{t}}(Q^d) = \chi(Q^d) \exp\left(\sum_i t_i (D_i \cdot d)\right). \quad (4.6)$$

This map from  $\Lambda_T$  to  $A[[\mathbf{t}]]$  is injective. To prove this, take a nonzero series. Properness in (4.5) makes its lowest-valuation support finite. Its initial image is a finite combination of distinct exponential characters. The finitely many exponent vectors are distinct integral vectors. Choose  $a = (a_i) \in \mathbf{Z}^\rho$  outside the finitely many integral hyperplanes on which two dot products agree, and substitute  $t_i = a_i s$ . The resulting exponents are distinct integers. Their derivatives at  $s = 0$  form a Vandermonde matrix with nonzero integer determinant, which remains invertible in the characteristic-zero fraction field. The characters are therefore linearly independent. The domain hypothesis on the associated graded prevents cancellation of the initial terms.

Let  $k_\chi = \overline{\text{Frac } A}$ . After embedding the two fraction fields in a common algebraic closure, injectivity makes the  $\chi_{\mathbf{t}}$ -module a scalar extension of the intrinsic module. On the other hand, the divisor equation identifies (4.6) with the degree-two formal bulk pullback of the  $\chi$ -module over  $B = k_\chi[[\mathbf{t}]]$ . Apply the finite-quotient argument in Lemma 4.4 with  $\eta = \sum_i t_i D_i$ : its integral- $z$  bulk gauge removes the positive divisor component. Hence vanishing over the intrinsic field implies vanishing after  $\chi$ .  $\square$

**Proposition 4.9** (Low-dimensional vanishing). *Let  $T$  be a point, a smooth projective curve, or a smooth projective surface. For every strictly Novikov-admissible specialization  $\chi$ ,*

$$\nu_6(T; \chi) = 0. \quad (4.7)$$

*Proof.* We first prove the intrinsic generic statement. If  $K_T$  is nef, the proof of [13, Claim 6.15] applies coefficient by coefficient over the small numerical Novikov ring. Set

$$\text{Gr} = \frac{\text{Deg} - \dim T}{2}, \quad g = \text{Gr} + \frac{1}{2}E,$$

where  $E$  is the identity in cohomological degrees of parity opposite to  $\dim T$  and is zero in the other degrees. The operator  $g$  has integral eigenvalues. Coefficientwise in  $Q^\beta$ , multiplication by  $c_1(T)$  sends

$$H^i(T) \longrightarrow H^{i+2-2c_1(T)\cdot\beta}(T).$$

If  $K_T$  is nef, then  $c_1(T) \cdot \beta \leq 0$ , so the Gr-weight rises by  $1 - c_1(T) \cdot \beta \geq 1$ . Conjugation by  $z^g$  therefore turns  $\nabla + \frac{1}{2}E dz/z$  into a regular-singular connection with nilpotent residue. Undoing the half-parity correction permits only the monodromy eigenvalues 1 and  $-1$ , never a primitive sixth root.

This treats curves of positive genus and minimal surfaces with nef canonical class. A point is immediate. The remaining minimal varieties are  $\mathbf{P}^1$ ,  $\mathbf{P}^2$ , and ruled surfaces; (4.2) reduces them to a point or a curve. Every nonminimal surface is obtained from a minimal surface by point blowups, and (4.3) adds only point summands. Hence the intrinsic generic invariant vanishes in every case. Lemma 4.8 gives (4.7).  $\square$

**Theorem 4.10** (Birational invariance through dimension four). *If  $Y$  and  $Y'$  are birational smooth projective complex varieties of the same dimension at most four, then*

$$\nu_6(Y) = \nu_6(Y'). \quad (4.8)$$

*Consequently, for smooth projective threefolds,*

$$Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1 \implies \nu_6(Y) = \nu_6(Y').$$

*Proof.* Weak factorization decomposes a birational map between smooth varieties into blowups and blowdowns with smooth centers of codimension at least two [1]. In dimension at most four every nontrivial center has dimension at most two. The calculation following Definition 4.7 shows that every center specialization in (4.3) is strictly admissible. Proposition 4.9 therefore makes every center term zero, so the endpoint values agree at each step.

If  $Y, Y'$  are smooth threefolds and  $Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1$ , apply the first assertion in dimension four and (4.2):

$$2\nu_6(Y) = \nu_6(Y \times \mathbf{P}^1) = \nu_6(Y' \times \mathbf{P}^1) = 2\nu_6(Y').$$

Cancelling two in  $\mathbf{Z}$  proves the stabilization assertion.  $\square$

**Proposition 4.11** (The cubic packet). *For every smooth cubic threefold  $X$ ,*

$$\nu_6(X) = 2. \quad (4.9)$$

*Proof.* In the basis  $1, P, P^2, P^3$ , Cai explicitly block-diagonalizes the small even quantum connection. The rank-two zero-exponential block has indicial polynomial

$$\rho^2 + \rho + \frac{5}{36},$$

with roots  $-1/6$  and  $-5/6$  [4, Section 3 and Proposition 6]. His gauge uses only integral powers of  $z$ . The two corresponding framed-monodromy eigenvalues are  $e^{-\pi i/3}$  and  $e^{\pi i/3}$ . The two remaining rank-one even blocks have formal-monodromy factor 1, hence contribute no primitive sixth root. Thus the multiplicity is exactly two. By definition,  $\nu_6$  uses the small even connection, so no odd-cohomology block enters this count. The numerical curve lattice of a cubic threefold has rank one, so the numerical-Novikov convention does not change the calculation.  $\square$

*Proof of Theorem 1.2.* The projective-bundle formula and Proposition 4.11 give

$$\nu_6(X \times \mathbf{P}^1) = 2\nu_6(X) > 0. \quad (4.10)$$

Suppose that  $X \times \mathbf{P}^1$  were rational. By Theorem 4.10,

$$\nu_6(X \times \mathbf{P}^1) = \nu_6(\mathbf{P}^4) = 0,$$

where the last equality follows from (4.2) over a point. This contradicts (4.10).  $\square$

#### 4.1 A genus-eight Fano consequence

Kuznetsov associates to every smooth prime Fano threefold  $V$  of genus eight a smooth Pfaffian cubic threefold  $X$ , together with rank-two vector bundles  $U$  on  $V$  and  $E^*$  on  $X$ , whose projectivizations are related by a flop [14, Theorems 2.17–2.18].

**Corollary 4.12** (One stabilization of  $V_{14}$ ). *For every smooth prime Fano threefold  $V$  of genus eight,*

$$\nu_6(V) = 2, \quad V \times \mathbf{P}^1 \text{ is irrational.} \quad (4.11)$$

*Proof.* The flop gives a birational map between the smooth projective fourfolds  $\mathbf{P}_V(U)$  and  $\mathbf{P}_X(E^*)$ . By Theorem 4.10, Proposition 4.6, and Proposition 4.11,

$$2\nu_6(V) = \nu_6(\mathbf{P}_V(U)) = \nu_6(\mathbf{P}_X(E^*)) = 2\nu_6(X) = 4.$$

Hence  $\nu_6(V) = 2$ . Formula (4.2) now gives  $\nu_6(V \times \mathbf{P}^1) = 4$ , whereas every rational smooth fourfold has  $\nu_6 = 0$  by Theorem 4.10.  $\square$

The argument is sharp for this invariant. In a factorization of the fivefold  $X \times \mathbf{P}^2$ , a center may have dimension three and may itself carry the cubic packet. The next stabilization therefore requires information about the packet's Stokes or Rees placement, not only its multiplicity.

## 5 The separation theorem

*Proof of Theorem 1.1.* Corollary 3.6 gives universal  $CH_0$ -triviality of  $X_b$ . The projective bundle formula for Chow groups gives the same property for  $X_b \times \mathbf{P}^1$ . Theorem 1.2 proves that this fourfold is irrational. Non-isotriviality follows from the nonconstant intermediate-Jacobian period map on the  $A_5$ -component: Hartlieb proves that its image is a one-dimensional special subvariety [9, Proposition 5.7].  $\square$



The algebraic class  $\Theta^4/4!$  removes the decomposition-of-diagonal obstruction, while the primitive-sixth framed-monodromy multiplicity survives one stabilization and obstructs a rational factorization. The  $A_5$ -family is where these conclusions hold simultaneously.

The two detectors are independent already on smooth projective threefolds. If  $U(Y)$  is the indicator of universal  $CH_0$ -triviality, then

$$(U(Y), \nu_6(Y)) = \begin{cases} (1, 0), & Y = \mathbf{P}^3, \\ (1, 2), & Y \text{ is a smooth member of the family above,} \\ (0, 2), & Y \text{ is a very general cubic threefold.} \end{cases}$$

The last line follows from the evenness theorem for algebraic curve classes on the very general intermediate Jacobian together with Voisin's criterion [7, Theorem 1.3 and Corollary 1.4][18, Corollary 4.4]. Thus neither universal  $CH_0$ -triviality nor  $\nu_6$  determines the other.

For  $X \times \mathbf{P}^m$  with  $m \geq 2$ , weak-factorization centers may have dimension at least three and can carry primitive-sixth quantum packets. The low-dimensional vanishing in Proposition 4.9 no longer proves noncancellation. Likewise, the fibrewise minimal-class argument gives no relative universal cycle over the pencil. These are separate problems and are not implicit in the theorem above.

Vanishing of the decomposition-of-the-diagonal obstruction therefore does not predict rationality after one explicit stabilization.

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